

Name:

University ID:

Please, answer the theoretical questions in detail, using only the provided sheets. The exam is **open-book**, and you can use **no electronic devices** except for a portable calculator. All questions and exercises will be corrected extensively; you are encouraged to expand beyond the strict topic of the question (without losing track of the target), like in an oral exam. You are also encouraged to draw sketches for concepts that are visual in nature.

The exam duration is **1h30m**.

This written exam is valid for 32 points. The lab assignments provide up to extra 4 points. Evaluations beyond 30/36 will be graded with 30 with honors.

Question 1

- A) Describe the concept of padding and stride in a convolution layer and why and how they are used. It is required to go into detail and provide a graphical representation of the concepts. Describe the advantages and disadvantages of the depth-wise + point-wise convolutions over the standard convolutions. It is required to be as detailed as possible.
- B) Given the function $f(x, y) = (x - 2)^2 + (y + 3)^2$, evaluate the gradient at $(x, y) = (1, -2)$.
- C) Assume training a neural network for 1000 epochs. Backpropagation requires $(0.003 * N_{\text{sample}} + 0.8)$ seconds per epoch. Assume a training set with 200000 samples and a sampling ratio of 34% for the stochastic selection of the samples. Compute total training time.

A) DW vs PW

1) MACs

2) WEIGHTS

1) CONV2D $\Rightarrow N_{\text{MAC}}^{\text{CONV}} = C_o C_{in} H_k W_k H_o W_o$

DW + PW $\Rightarrow N_{\text{MAC}}^{\text{DP}} = (C_{in} H_k W_k + C_o C_{in}) H_o W_o$

2) CONV2D $\Rightarrow N_{\text{WEIGHTS}}^{\text{CONV}} = C_o C_{in} H_k W_k$

DW + PW $\Rightarrow N_{\text{WEIGHTS}}^{\text{DP}} = C_{in} H_k W_k + C_o C_{in}$

$\Rightarrow (DW + PW) \ll CONV2D$

Parameters: $C_o = 6, H_k = W_k = 3, C_{in} = 8, H_o = W_o = 16$

Calculations:

- $N_{\text{MAC}}^{\text{CONV}} = 147456$
- $N_{\text{MAC}}^{\text{DP}} = 34816$
- $N_{\text{WEIGHTS}}^{\text{CONV}} = 576$
- $N_{\text{WEIGHTS}}^{\text{DP}} = 136$

B)

$\frac{\partial f}{\partial x} = 2(x - 2) \Rightarrow \frac{\partial f}{\partial x}(1, -2) = -2$

$\frac{\partial f}{\partial y} = 2(y + 3) \Rightarrow \frac{\partial f}{\partial y}(1, -2) = 2$

C)

$T_{\text{BP}} = 0.003 * N_{\text{sample}} + 0.8 \text{ s/epoch}$

$N_{\text{sample}} = 200000 * 0.34 = 68000$

$T_{\text{BP}} = 0.003 * 68000 + 0.8 = 204.8 \text{ s}$

$T_{\text{TRAIN}} = T_{\text{BP}} * 1000 = 204800 \text{ s}$

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Question 2

A parallel cluster of 8 RISC-V processors executed a convolutional layer of 3×3 64 filters applied on an input feature map of 40×40 pixels over 32 input channels.

Inputs are quantized to 8-bits, while weights are 4-bits.

It is applied padding with zeros on the input, and stride=1. There are no bias values.

At the output of the convolutional layer, the RELU non-linearity is applied.

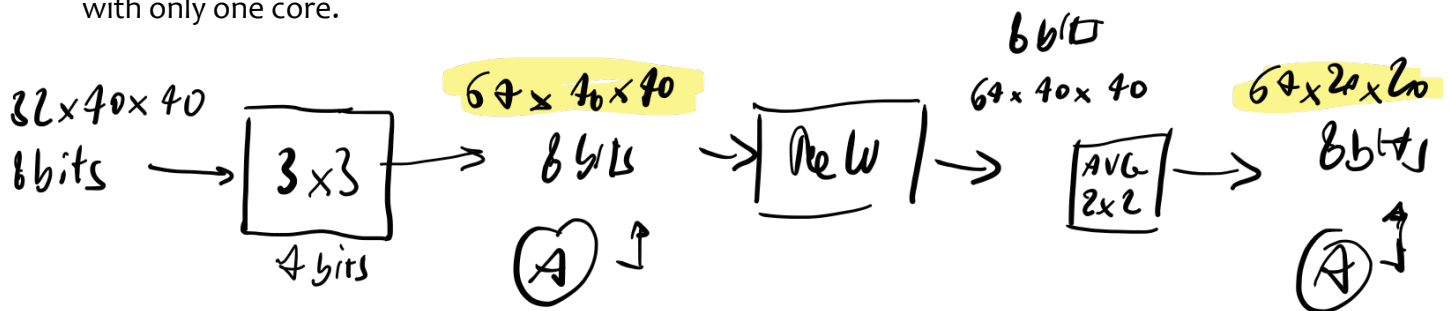
After the RELU, the output feature map is average-pooled with a 2×2 pooling window.

Assume the following performance of each RISC-V processor:

- One MAC operation performed in 3 clock cycles;
- One ADD operation performed in 3 clock cycles;
- One RELU operation performed in 5 clock cycles;
- One division operation performed in 10 clock cycles;

Assume that the three kernels (convolution, ReLU, average pooling) are parallelized over the output channels with no parallelization overhead.

- Compute the size of the tensor (H, W, OCH) at the output of the convolutional layer. Then, compute the size of the tensor (H_P , W_P , OCH_P) at the output of the pooling layer.
- Compute the amount of memory needed to store the complete layer (inputs, outputs, weights, intermediate activations)
- Compute the total number of cycles to process the layer on the 8-core cluster.
- Compute the speed-up obtained by processing the layer in parallel with 8 cores versus sequentially with only one core.



B) IN: $32 \times 40 \times 40 \times (8 \text{ bits}) = 409600 \text{ b} = 51.200 \text{ B}$
 WGT: $32 \times 64 \times 3 \times 3 \times (4 \text{ bits}) = 73728 \text{ b} = 9216 \text{ B}$
 OUT_{conv}: $64 \times 40 \times 40 \times (8 \text{ bits}) = 819200 \text{ b} = 102400 \text{ B}$
 OUT_{ReLU}: $64 \times 40 \times 40 \times (8 \text{ bits}) = \quad //$
 OUT_{AVG}: $64 \times 20 \times 20 \times (8 \text{ bits}) = 204800 \text{ b} = 25600 \text{ B}$
 TOT: $2326528 \text{ b} = 290816 \text{ B}$

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$$\begin{aligned} c) \quad & \text{CON} + \text{ReLU} + \text{AVG} \\ \Rightarrow & 32 \times 64 \times 40 \times 40 \times 3 \times 3 \text{ MAC} + \\ & + 64 \times 40 \times 40 \text{ ReLU} \\ & + 64 \times 20 \times 20 (3 \text{ ADD} + 1 \text{ DIV}) = \\ = & 29\,491\,200 \text{ MAC} + 102\,400 \text{ ReLU} \\ & + 25\,600 (3 \text{ ADD} + 1 \text{ DIV}) \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Total ops} &= 88\,473\,600 + 512\,000 + \\ &+ 486\,400 = 89\,472\,000 \end{aligned}$$

d) SU?

$$\begin{aligned} \text{Ops}_{\text{SU}}^{\text{bc}} &= 88\,473\,600/b + 512\,000/b \\ &+ 486\,400/b = 11\,184\,000 \end{aligned}$$

$$\Rightarrow \text{SU} = b$$

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Question 3

Consider the weight matrix in the figure. Quantize the matrix targeting a linear affine quantization in int8.

-1.08	0.78	2.01
-0.54	0.21	0.3
0.15	-1.0	1.0

$$r = S(q - Z)$$

REAL - INT TRANSF FORMULA

$$S = \frac{r_{\max} - r_{\min}}{q_{\max} - q_{\min}}$$

SCALE

$$Z = \text{round} \left(q_{\min} - \frac{r_{\min}}{S} \right)$$

ZERO - POINT

In particular:

- Compute the scale factor (S) and the zero point (Z).
- Compute the quantized matrix in int8.
- Discuss other quantization strategies, including non-uniform quantization and weights clustering. Regarding non-uniform quantization, it is also necessary to discuss log-domain quantization and use cases where this method can be advantageous. Regarding weights clustering, you MUST ALSO answer and comment on this question: "Can the weights clustering be applied to input activations? If yes/no, why?"

8bits :

$$\begin{aligned} r_{\max} &= 2.01 & q_{\min} &= -2^7 = -128 \\ r_{\min} &= -1.08 & q_{\max} &= 2^7 - 1 = 127 \end{aligned}$$

$$S = \frac{2.01 + 1.08}{127 + 128} = \frac{3.09}{255} = 0.0121$$

$$Z = \text{round} \left(-128 + \frac{1.08}{0.0121} \right) = \text{round}(-38.74) = -39$$

Bit Width	q _{min}	q _{max}
2	-2	1
3	-4	3
4	-8	7
N	-2 ^{N-1}	2 ^{N-1} -1

$$q = \text{round} \left(\frac{r}{S} + Z \right)$$

-1.08	0.78	2.01
-0.54	0.21	0.3
0.15	-1.0	1.0

-128	25	127
-84	22	-14
-27	-12	44

